

## Chapter 2, Section 7

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1. Let  $(X, \mathcal{O}_X)$  be a locally ringed space, and let  $f : \mathcal{L} \rightarrow \mathcal{M}$  be a surjective map of invertible sheaves on  $X$ . Show that  $f$  is an isomorphism.

*Proof.* We reduce to the following algebraic problem: Any  $A$ -linear surjective map  $\phi : A \rightarrow A$  is an isomorphism. Indeed,  $\phi$  is determined by  $r = \phi(1)$ , and  $\phi$  surjective implies there exists  $a \in A$  such that  $ar = \phi(a) = 1$ . Thus,  $r$  is a unit, and  $\phi$  is injective.  $\square$

2. Let  $X$  be a scheme over a field  $k$ . Let  $\mathcal{L}$  be an invertible sheaf on  $X$ , and let  $\{s_0, \dots, s_n\}$  and  $\{t_0, \dots, t_m\}$  be two sets of sections of  $\mathcal{L}$ , which generate the same subspace  $V \subseteq \Gamma(X, \mathcal{L})$ , and which generate the sheaf  $\mathcal{L}$  at every point. Suppose  $n \leq m$ . Show that the corresponding morphisms  $\varphi : X \rightarrow \mathbf{P}_k^n$  and  $\psi : X \rightarrow \mathbf{P}_k^m$  differ by a suitable linear projection  $\mathbf{P}^m - L \rightarrow \mathbf{P}^n$  and an automorphism of  $\mathbf{P}^n$ , where  $L$  is a linear subspace of  $\mathbf{P}^m$  of dimension  $m - n - 1$ .

3. Let  $\varphi : \mathbf{P}_k^n \rightarrow \mathbf{P}_k^m$  be a morphism. Then:

- (a) either  $\varphi(\mathbf{P}^n) = \mathbf{P}^m$  or  $m \geq n$  and  $\dim \varphi(\mathbf{P}^n) = n$ ;
- (b) in the second case,  $\varphi$  can be obtained as the composition of (1) a  $d$ -uple embedding  $\mathbf{P}^n \rightarrow \mathbf{P}^N$  for a uniquely determined  $d \geq 1$ , (2) a linear projection  $\mathbf{P}^N - L \rightarrow \mathbf{P}^m$ , and an automorphism of  $\mathbf{P}^m$ . Also,  $\varphi$  has finite fibers.

*Proof.*

- (a) Let  $\mathcal{L}$  be the invertible sheaf on  $\mathbf{P}^n$  associated to the morphism  $\varphi : \mathbf{P}^n \rightarrow \mathbf{P}^m$ , which is generated by global sections  $s_i = \varphi^*(x_i)$ ,  $i = 0, 1, \dots, m$ . Pick  $P \in \varphi^{-1}((1, 0, \dots, 0))$  (we can assume such  $P$  always exists by applying an automorphism of  $\mathbf{P}^m$ ). Identifying  $\mathcal{L}_P \cong \mathcal{O}_{P, \mathbf{P}^n}$ , we see that  $(s_i)_P$  for  $i \neq 0$  is not a unit, and they generate the local ring of  $\mathcal{O}_{P, \mathbf{P}^n}$ . Since  $\mathbf{P}^n$  is smooth, the minimal number of generators of the maximal ideal of  $\mathcal{O}_{P, \mathbf{P}^n}$  is equal to the dimension  $n$ . The sections  $s_i$  generate  $\mathcal{O}_{P, \mathbf{P}^n}$ . Hence,  $m \geq n$ . Our proof also shows the induced homomorphism of local rings  $\varphi_P : \mathcal{O}_{\varphi(P), \mathbf{P}^m} \rightarrow \mathcal{O}_{P, \mathbf{P}^n}$  is surjective, which implies  $\mathcal{O}_{\varphi(P), \varphi(\mathbf{P}^n)} = \mathcal{O}_{\varphi(P), \mathbf{P}^m} / \ker \varphi_P \cong \mathcal{O}_{P, \mathbf{P}^n}$ . Hence,  $\varphi(P)$  has dimension  $n$ .  $\square$

5. Establish the following properties of ample and very ample invertible sheaves on a noetherian scheme  $X$ .  $\mathcal{L}, \mathcal{M}$  will denote invertible sheaves, and for (d), (e) we assume furthermore that  $X$  is of finite type over a noetherian ring  $A$ .

- (a) If  $\mathcal{L}$  is ample and  $\mathcal{M}$  is generated by global sections, then  $\mathcal{L} \otimes \mathcal{M}$  is ample.
- (b) If  $\mathcal{L}$  is ample and  $\mathcal{M}$  is arbitrary, then  $\mathcal{M} \otimes \mathcal{L}^n$  is ample for sufficiently large  $n$ .
- (c) If  $\mathcal{L}, \mathcal{M}$  are both ample, so is  $\mathcal{L} \otimes \mathcal{M}$ .
- (d) If  $\mathcal{L}$  is very ample and  $\mathcal{M}$  is generated by global sections, then  $\mathcal{L} \otimes \mathcal{M}$  is very ample.
- (e) If  $\mathcal{L}$  is ample, then there is an  $n_0 > 0$  such that  $\mathcal{L}^n$  is very ample for all  $n \geq n_0$ .

*Proof.*

- (a) Fix a coherent sheaf  $\mathcal{F}$  on  $X$  for the remainder of this problem. Let  $n \gg 0$  such that  $\mathcal{F} \otimes \mathcal{L}^n$  is generated by global sections. If  $\mathcal{M}$  is generated by global sections, then so is  $\mathcal{M}^n$  for any  $n > 0$ . More generally, the tensor product of two invertible sheaves generated by global sections is generated by global sections. Indeed, if  $t_i$  and  $r_j$  are generating global sections of  $\mathcal{L}$  and  $\mathcal{M}$ , then  $\mathcal{L} \otimes \mathcal{M}$  is generated by  $t_i \otimes r_j$  for all pairs  $(i, j)$  (tensor products commutes with colimits). Thus,  $\mathcal{F} \otimes \mathcal{L}^n \otimes \mathcal{M}^n \cong \mathcal{F} \otimes (\mathcal{L} \otimes \mathcal{M})^n$  is generated by global sections. Hence,  $\mathcal{L} \otimes \mathcal{M}$  is ample.
- (b) Let  $n > 0$  such that  $\mathcal{M} \otimes \mathcal{L}^{n-1}$  is generated by global sections. By (a),  $\mathcal{L} \otimes (\mathcal{M} \otimes \mathcal{L}^{n-1}) = \mathcal{M} \otimes \mathcal{L}^n$  is ample.
- (c) If  $\mathcal{M}$  is ample, then  $\mathcal{M}^n$  is generated by global sections for large enough  $n$ . Also,  $\mathcal{L}^n$  is ample by (II, 7.5), so  $\mathcal{L}^n \otimes \mathcal{M}^n$  is ample. Hence,  $\mathcal{L} \otimes \mathcal{M}$  is ample by (II, 7.5) again.

- (d) Let  $i : X \rightarrow \mathbf{P}_A^n$  be an immersion such that  $\mathcal{L} \cong i^* \mathcal{O}_{\mathbf{P}_A^n}(1)$ , and let  $\varphi : X \rightarrow \mathbf{P}_A^m$  be the unique  $A$ -morphism corresponding to  $\mathcal{M}$  (II, 7.1). By the universal property of the fiber product  $\mathbf{P}_A^n \times_A \mathbf{P}_A^m$ , there exists a unique  $A$ -morphism  $\phi : X \rightarrow \mathbf{P}_A^n \times_A \mathbf{P}_A^m$  such that  $i = p_n \circ \phi$  and  $\varphi = p_m \circ \phi$ , where  $p_n : \mathbf{P}_A^n \times_A \mathbf{P}_A^m \rightarrow \mathbf{P}_A^n$  and  $p_m : \mathbf{P}_A^n \times_A \mathbf{P}_A^m \rightarrow \mathbf{P}_A^m$  are the natural projection maps. It is not hard to see  $\phi$  is an immersion, and composing it with a Segre embedding (Ex. 5.12), we obtain an immersion  $\phi' : X \rightarrow \mathbf{P}_A^N$ , where  $N = nm + n + m$ , such that  $\mathcal{L} \otimes \mathcal{M} \cong \phi'^* \mathcal{O}(1)$ .
- (e) If  $\mathcal{L}$  is ample, then there exists  $n_0 > 0$  such that  $\mathcal{L}^{n-1}$  is generated by global sections for all  $n \geq n_0$ . By (d),  $\mathcal{L}^n = \mathcal{L} \otimes \mathcal{L}^{n-1}$  is very ample for all  $n \geq n_0$ .

□